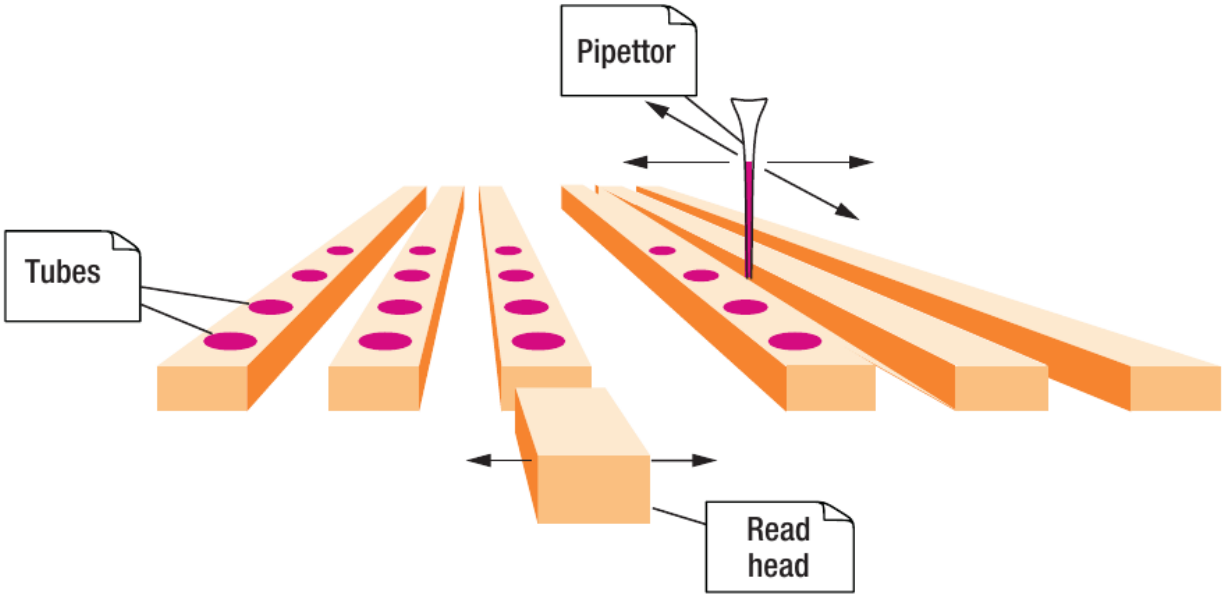
Online scheduling in a biological analysis system (1/5)

- An electromechanical system for immunoenzymatic analysis by BioMérieux
  - Each slot comprises several test tubes
  - A pipettor and a read head can act simultaneously on test tubes

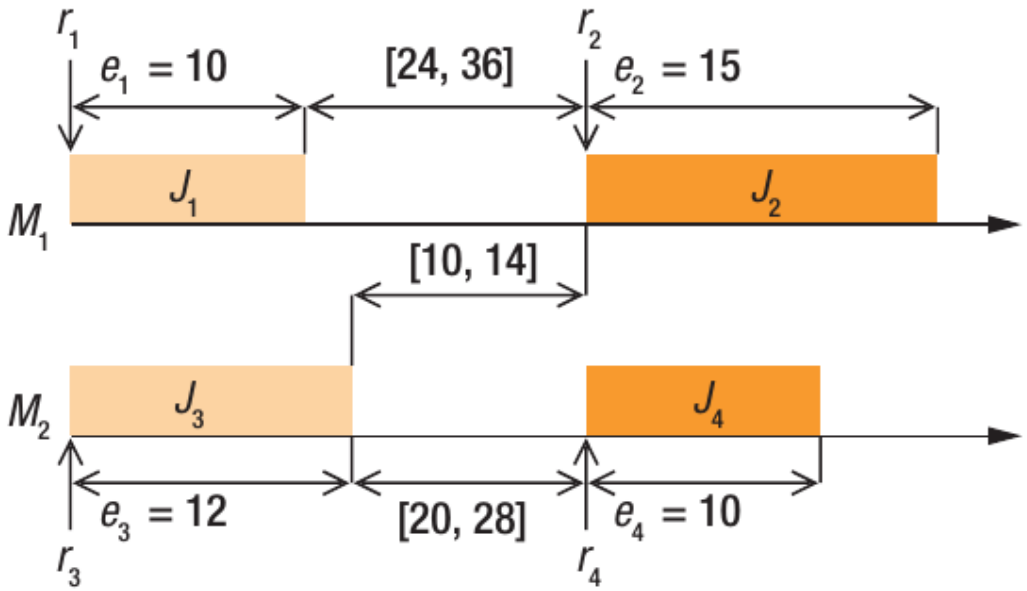


L. Ridi, J. Torrini, E. Vicario, "Developing a Scheduler with Difference-Bound Matrices and the Floyd-Warshall Algorithm," IEEE Software, 2012.

E. Vicario, L. Ridi, A. Carignano, J. Torrini, "Job scheduler for electromechanical system for biological analyses," Patent WO2013098088A1, 2013.

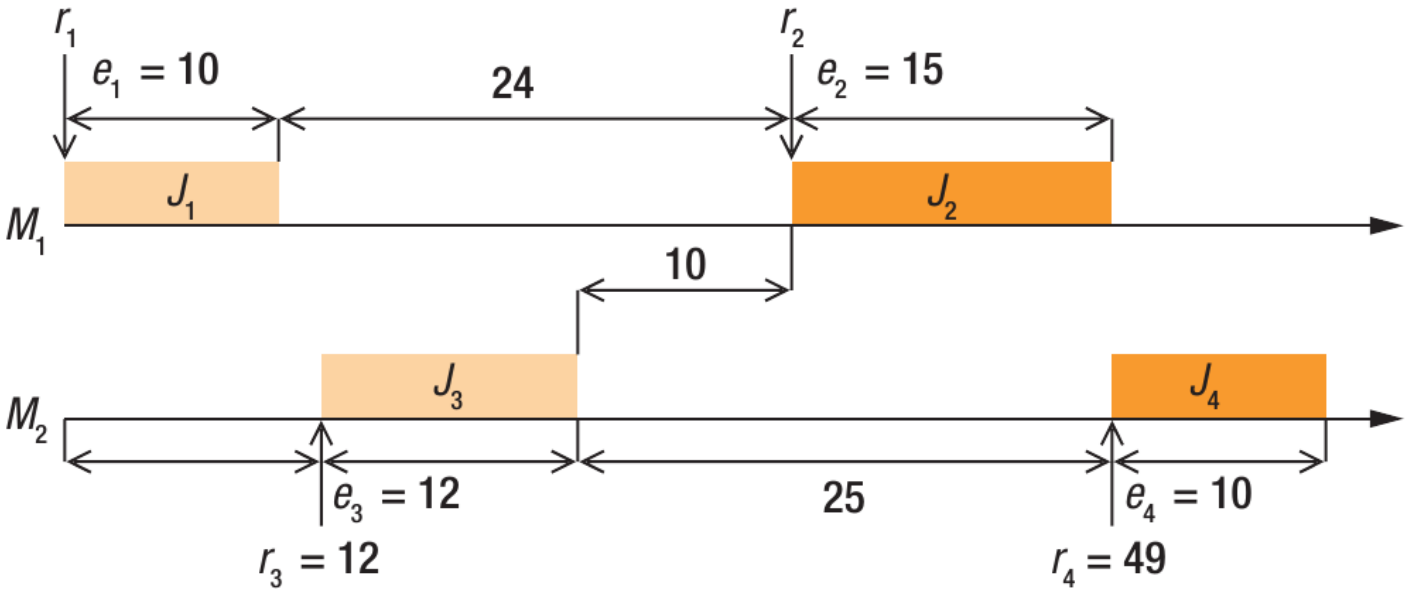
Online scheduling in a biological analysis system (2/5)

- An example of the scheduling problem
  - 4 jobs J_1, J_2, J_3, J_4 allocated on two machines M_1, M_2 → TESTINA E PIPETTA
  - e indicates an execution time, r indicates a release time
  - J_2 has a mutual exclusion with J_4
  - J_2 has a precedence constraints wrt the completion of J_1 and J_3
  - J_4 has a precedence constraints wrt the completion of J_3



Online scheduling in a biological analysis system (3/5)

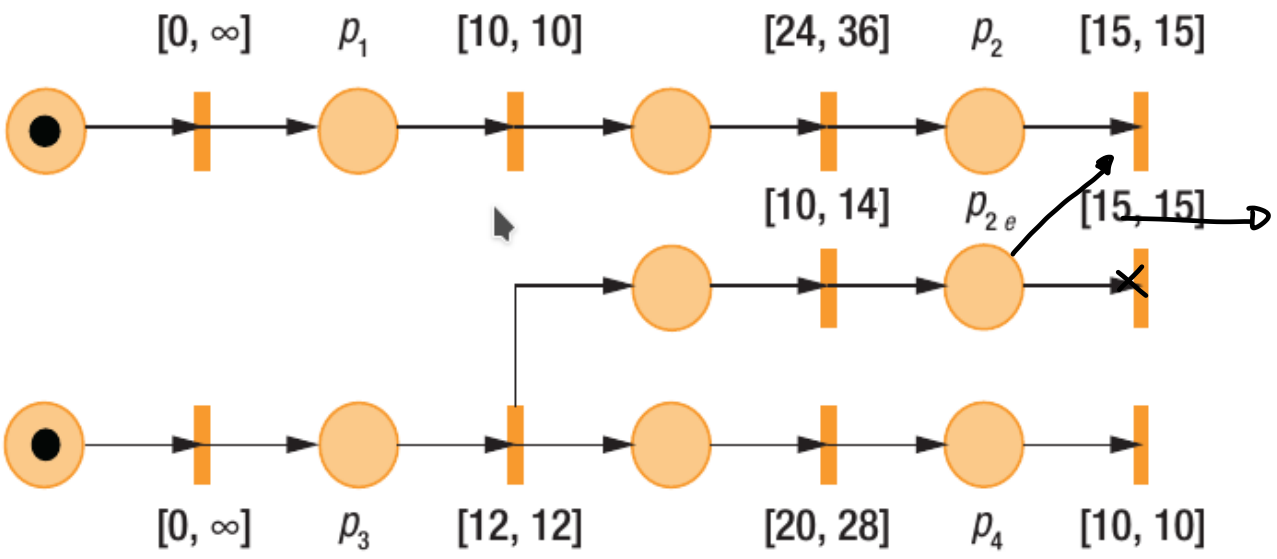
- A schedule that minimizes the overall completion time



- How to compute such a schedule?

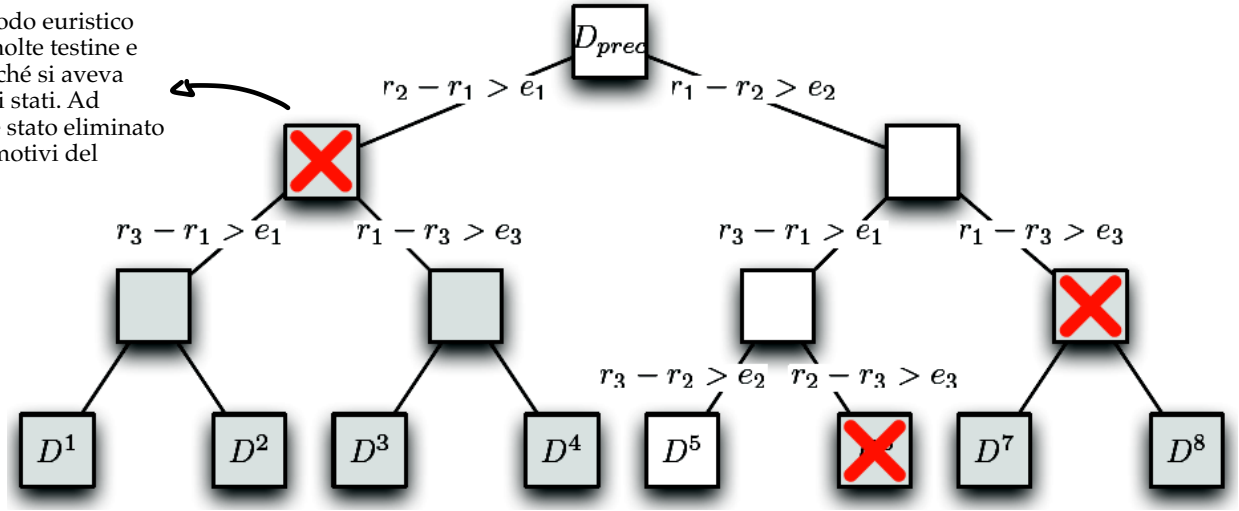
Online scheduling in a biological analysis system (4/5)

- TPN encoding sequencing and timing constraints of the scheduling problem
- An execution is correct if and only if it never visits a state class where: = Non si verificano mai le seguenti:
  - place p_2 has a token while place p_{2e} is empty (O VICEVERSA)
  - both places p_2 and p_4 contain a token — PERCHÉ J2 E J4 SONO IN MUTEX



Funziona anche come segnato, solo che nell'altro modo quando si va a fare il grafo delle classi degli stati compaiono anche i percorsi non permessi (quelli dov'è queste due transizioni non sparano insieme) e si possono eliminare. In pratica le due soluzioni sono quindi le stesse ma l'altra è più completa.

Online scheduling in a biological analysis system (5/5)

- State class graph enumeration to derive a schedule
- Optimization: heuristic-based enumeration of the state class graph
- Tests on challenging cases (e.g., 24 pipettor actions, 8 read head actions)
- Faster compared to traditional model checking approaches
- Software implementation running on a real system!

Sì sono esclusi percorsi in modo euristico per via di troppe variabili (molte testine e molte provette) e quindi perché si aveva un'esplosione del grafo degli stati. Ad esempio un ramo del grafo é stato eliminato perché non era fattibile per motivi del protocollo biologico.